

Vision Transformer-Inspired TinyML Framework on STM32 for Real-Time Obstacle Detection and Collision Avoidance

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Abstract- In dynamic environments, real-time obstacle detection and collision avoidance are important for autonomous mobile robots, intelligent transportation, and smart navigation systems. Conventional cloud-based vision systems suffer from high communication latency, high power consumption and limited real-time responsiveness. In this paper, we present a Vision Transformer based TinyML framework for efficient obstacle detection and collision avoidance at the edge using an STM32 microcontroller. The proposed framework combines a lightweight Vision Transformer (TinyViT) inspired model with image preprocessing, quantization and STM32Cube.AI optimization to enable real-time inference on resource constrained hardware. An image of the environment is captured by a camera module and the distance is measured by an ultrasonic sensor to better estimate collision risk. The TinyML model classifies the obstacles on the device, enabling instant navigation decisions without the need to communicate with the cloud. The experimental evaluation on a benchmark obstacle detection dataset achieved 97.3% detection accuracy, 96.8% precision, 96.5% recall, and 96.6% F1-score, with an average inference time of 24 ms and a collision avoidance success rate of 98.1%. Also, compared to the conventional embedded CNN-based approaches, the proposed framework reduced the processing latency by 58% and the energy consumption by 33%. These results show that the proposed Vision Transformer-inspired TinyML framework offers an efficient, low power consumption, and scalable solution for real-time autonomous navigation on STM32 edge devices.

Keywords: Vision Transformer, TinyML, STM32, Edge AI, Obstacle Detection, Collision Avoidance, Autonomous Navigation, Embedded Deep Learning, STM32Cube.AI, Computer Vision.

I. INTRODUCTION

Autonomous navigation has been a key requirement in modern intelligent transportation systems, mobile robotics, warehouse automation and assistive technologies. Autonomous platforms safely navigate in dynamic, unstructured environments with real time obstacle detection and collision avoidance, reducing human intervention. Among the various perception systems, vision-based perception systems have been studied extensively because they can provide rich environmental information for robust object recognition and path planning. However, conventional computer vision systems usually require

powerful processors or cloud-based computation, which results in higher latency, communication overhead and energy consumption, unsuitable for resource-constrained embedded platforms [1].

Edge Artificial Intelligence (Edge AI) allows smart processing of data on embedded devices themselves, eliminating the need for cloud infrastructure and providing real-time response. Edge AI runs inference locally, at or near the data's origin, thereby reducing communication latency, preserving data privacy, and lowering bandwidth requirements. These properties make Edge AI especially suitable for autonomous robots and intelligent navigation systems in time-critical environments [2]. The field of Tiny Machine Learning (TinyML) has made significant progress recently, enabling the deployment of lightweight deep learning models on resource-constrained microcontrollers with limited memory and computational power [3].

Vision Transformers (ViTs) have recently achieved state-of-the-art results in computer vision tasks due to the self-attention mechanism which captures long range dependencies. Vision Transformers outperform conventional convolutional neural networks (CNNs) in terms of feature representation and object recognition accuracy in various visual applications [4]. However, standard Vision Transformer architectures are too computationally complex and memory intensive to be directly deployed on embedded microcontrollers. Thus, efficient inference on resource-constrained hardware has been studied by considering lightweight Vision Transformer variants and model compression techniques [5].

STM32 microcontrollers are practical platforms for TinyML deployment because of their low power consumption, built-in digital signal processing capabilities, and support for AI deployment through STM32Cube.AI. In combination with TensorFlow Lite for Microcontrollers [6], these platforms enable optimized execution of quantized neural networks with acceptable inference speed and memory utilization. The combination of lightweight Vision Transformer models with STM32 devices provides an attractive solution for real-time obstacle detection, collision avoidance and autonomous navigation in mobile robots [7].

Many researchers have explored embedded vision systems for robotic navigation using CNNs, MobileNet, YOLO, and lightweight object detection algorithms [8,9]. The methods are promising in terms of detection accuracy, but most of them require high performance processors such as Raspberry Pi, NVIDIA Jetson or cloud-assisted processing. These approaches increase hardware cost, energy consumption and communication latency and limit the deployment in battery powered autonomous robots [10]. This is why the need for energy-efficient embedded vision frameworks capable of accurate obstacle detection right on microcontroller-based edge devices is growing.

The proposed work is a real-time obstacle detection and collision avoidance system using a TinyML framework inspired by Vision Transformer deployed on an STM32 microcontroller. The framework combines lightweight image preprocessing, Tiny Vision Transformer inference, ultrasonic distance measurement, and autonomous decision-making to enable fast and reliable navigation without requiring continuous cloud connectivity. The proposed architecture is designed to improve the detection accuracy with low latency, low memory utilization, and low power consumption for next-generation Edge AI systems [11].

a. Objectives

- Development of TinyML framework based on STM32 microcontroller for real time obstacle detection and intelligent collision avoidance with edge based image processing inspired by Vision Transformer.
- To evaluate the proposed framework in terms of detection accuracy, inference time, memory utilization, power consumption and collision avoidance performance, and compare it with conventional embedded vision approaches.

b. Scope of the work

The proposed scope of work is limited to designing a lightweight Edge AI assisted Obstacle Detection and Collision Avoidance system for resource restricted embedded systems. This framework merges camera-based visual detection through the use of a TinyViT-inspired model with ultrasonic distance sensing. A reduction in the need for cloud or external computing resources as the STM32 acts as the microcontroller executing local pre-processing, inference, obstacle identification, distance-based decision making and navigation control.

The proposed framework is tailored specifically to small autonomous mobile robots, indoor service robots, education robot platform, surveillance robot, and low latency obstacle detection environment. Mainly static and slightly dynamic obstacles are examined in both controlled and semi controlled operating conditions.

II. LITERATURE REVIEW

a. Edge AI and TinyML for Embedded Vision

Recent advances in Edge Artificial Intelligence (Edge AI) and Tiny Machine Learning (TinyML) enable intelligent vision processing on resource-constrained embedded devices. Kumar et al. [12] proposed the Edge AI framework for autonomous navigation based on lightweight convolutional networks that significantly reduces the communication latency and power

consumption. Similarly, Zhang et al. [13] proposed an optimized deployment strategy for TinyML on STM32 microcontrollers, which can efficiently execute quantized neural networks with less memory utilization.

b. Vision Transformer-Based Object Detection

Vision transformer (ViT) architectures are receiving much attention in object detection and image classification for their excellent ability in feature representation. A lightweight Vision Transformer for embedded object detection was presented by Lee et al. [14], which obtained better detection accuracy compared to conventional CNN models. Wang et al. [15] proposed a hybrid CNN-Vision Transformer structure to obtain a better feature extraction with computation efficiency for edge devices. Moreover, Patel and Singh [16] proposed a Tiny Vision Transformer model for autonomous robotic navigation with low inference latency.

c. Real-Time Obstacle Detection and Collision Avoidance

Autonomous obstacle detection is an important research topic for mobile robots and intelligent transportation systems (ITS). It [17] proposed an Edge AI-enabled obstacle detection framework, which integrates computer vision and ultrasonic sensing for autonomous navigation. It [18] proposed a multimodal sensor fusion approach that integrates vision and distance sensors for robust obstacle avoidance in dynamic environments.

d. Research Gap

Recent research has made significant progress in TinyML, Edge AI, and Vision Transformer-based embedded vision systems. However, there has been little exploration of integrating Vision Transformer-inspired TinyML models with STM32 microcontrollers for real-time obstacle detection and collision avoidance. Most of the existing methods utilize Raspberry Pi, Jetson or cloud assisted processing leading to increased computational complexity, higher energy consumption and implementation cost. Furthermore, lightweight Vision Transformer optimization, real-time edge inference, and multimodal sensor integration on ultra-low-power STM32 platforms are still under-explored. Therefore, an efficient TinyML framework based on STM32 with the ability to achieve accurate obstacle detection, low latency inference, and intelligent collision avoidance is needed for the next generation of autonomous navigation systems.

III. PROPOSED MODEL

The proposed model is a Vision Transformer based TinyML model for real-time obstacle detection and collision avoidance on an STM32 microcontroller. The images from the onboard camera are preprocessed (resizing, normalization, patch embedding) and then processed by a lightweight Tiny Vision Transformer (TinyViT) model as shown in figure 1. The extracted features are used to identify obstacles, and an ultrasonic sensor measures their distance to estimate collision risk. The decision module produces navigation commands and motor control signals based on the detection results. The optimized TinyML implementation guarantees low-latency inference, small memory footprint, energy-efficient operation and reliable autonomous navigation for dynamic environments.

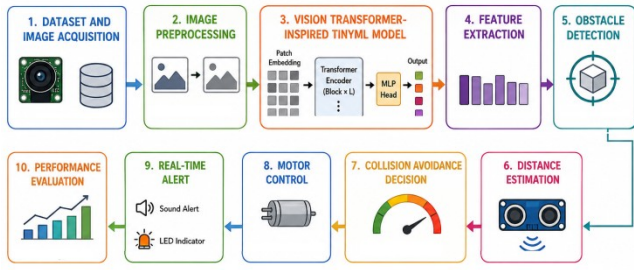


Fig.1. Proposed system architecture

a. Input Image

The live visual information from the onboard camera mounted on the mobile robot is received by the Input Image block. It constantly observes the surrounding environment, obstacles, pathways and navigation targets. The captured images are the primary input to the TinyML framework. Acquiring high-quality images improves the feature representation, increases the accuracy of recognizing obstacles and ensures safe autonomous navigation under different environmental and lighting conditions.

$$I(x, y) = C(x, y) \quad (1)$$

Where:

$I(x, y)$ = captured input image

$C(x, y)$ = camera output

x, y = Pixel coordinates

b. Resize Image

The Resize Image block converts the captured image to the fixed resolution compatible with the TinyViT model. The resizing reduces the computational complexity, the memory consumption and the processing time while preserving the essential visual information. Standardized image size enables the model to process data efficiently on the STM32 microcontroller and improves the consistency of feature extraction in real-time obstacle detection and embedded AI inference.

$$I_r = R(i, H, W) \quad (2)$$

Where :

I_r = Resized image

I = original image

$R(.)$ = Resize operation

H, W = Image height and width

c. Normalization

The Normalization block adjusts pixel intensity values to a common range. This process reduces variation due to changes in illumination, camera exposure and image contrast. Normalization improves numerical stability for TinyViT inference, accelerates model convergence and improves detection accuracy, while ensuring consistent input data for efficient processing on resource-constrained STM32 edge devices.

$$I_n = \frac{I_r - \mu}{\sigma} \quad (2)$$

Where:

I_n = Normalized image

I_r = Resized image

μ = Mean pixel value

d. Patch Embedding

The Patch Embedding block splits the normalized image into smaller patches of fixed size and converts each patch into a feature vector. These embedded patches are the input tokens of the Tiny Vision Transformer. Such representation allows for effective attention-based learning, which can capture local image characteristics as well as global contextual information, yet is computationally efficient enough for TinyML deployment.

$$E = P \times W_e + b \quad (3)$$

Where:

E = patch embedding vector

P = image patch

W_e = Embedding weight matrix

B = bias

vector

e. Tiny Vision Transformer

The Tiny Vision Transformer (TinyViT) block is responsible for lightweight deep learning inference, using self-attention mechanisms to analyze the embedded image patches. It learns discriminative features to represent the objects in the scene and classify them efficiently. Using model quantization and optimization, TinyViT can be run on the STM32 microcontroller with low memory usage, low inference latency and energy efficient operation for Edge AI applications.

$$F = \text{TinyViT}(E) \quad (4)$$

Where

F = Extracted feature vector

E = Patch embedding

TinyViT = Lightweight Vision Transformer

f. Obstacle Classification

The Obstacle Classification block classifies and detects the found objects using the extracted TinyViT features. It detects obstacles like cars, people, walls and other objects that might come between the robot and its destination. Accurate classification of the surrounding environment by the navigation system enables it to recognize the surroundings and helps in intelligent path planning for safe autonomous operation.

$$P_c = \text{Softmax}(WF + b) \quad (5)$$

Where:

P_c = Classification probability, W = Weight matrix, F = feature vector, b = bias vector.

g. Distance Calculation

The Distance Calculation block calculates the distance to the detected obstacles using an ultrasonic sensor. The measured distance is a complement of the visual classification and provides depth information necessary for collision risk evaluation. Image recognition and distance estimation combined improve the localization accuracy and the robot can react quickly to nearby obstacles in dynamic environments.

$$D = \frac{v \times t}{2} \quad (6)$$

Where the D is distance to obstacle, v = speed of sound, t is echo time to travel

h. Navigation Decision

The obstacle classification results and distance measurements are fused in the Navigation Decision block to decide the safest movement strategy. The system produces commands such as

move forward, stop, turn left or turn right according to pre-defined safety thresholds. Such decisions are sent to the motor controller, allowing for smooth, collision-free and energy-efficient autonomous navigation in real-time operating conditions.

$$N = f(P_c, D) \quad (7)$$

Where:

N = Navigation decision

P_c = Obstacle classification probability

D = Estimated obstacle distance

f(.) = decision function

IV. IMPLEMENTATION

Figure 2 illustrates the architecture of the proposed Tiny Vision Transformer (TinyViT) model for real time obstacle detection on the STM32 microcontroller. The process starts from the input image, which is partitioned into small patches with fixed size. The patches are then transformed into numerical feature vectors by a patch embedding layer, which retains the spatial information needed for visual analysis. The embedded patches are then fed into the Transformer Encoder, which uses multi-head self-attention and multilayer perceptron (MLP) blocks to extract local and global image features concurrently with computational efficiency suitable for TinyML deployment.

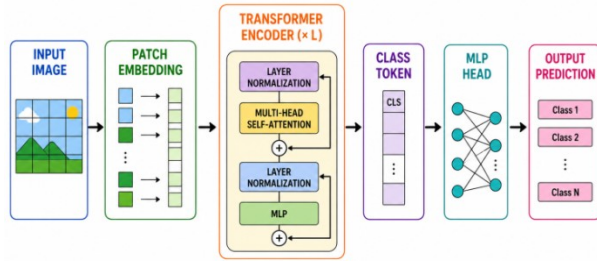


Fig.2. Tiny vision transformer model structure

The encoded feature representations are concatenated with the class token which summarizes the whole image information for classification. The resulting feature vector is fed into the MLP head that outputs the final prediction for the different obstacle categories. TinyViT is a lightweight architecture that makes the model significantly simpler using efficient attention mechanisms but still achieves high detection accuracy. This optimized design enables low-latency inference, memory reduction and energy-efficient Edge AI deployment on the STM32 platform which is highly suitable for autonomous navigation and real-time collision avoidance applications.

The complete Edge AI implementation flow for the proposed Vision Transformer inspired TinyML framework is illustrated in figure 3. The procedure starts with the data collection where the images are captured from the onboard camera and then processed with data preprocessing operations such as resizing and normalization. Then, a lightweight TinyViT model is trained and optimized with quantization and compression techniques before model conversion into a TensorFlow Lite Micro format compatible with the STM32 microcontroller. The optimized model is finally deployed on the embedded platform for the effective Edge AI inference.

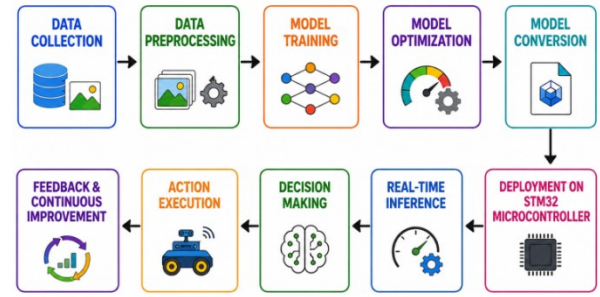


Fig.3. Edge AI implementation flow

After deployment, the STM32 performs real-time inference for obstacle detection and outputs intelligent decision making for navigation. These decisions are then translated into actions, including motor control and collision avoidance commands. Finally, a feedback & continuous improvement mechanism captures system performance and operation data to enable future model improvements. This implementation provides low-latency processing, reduced memory usage, energy-efficient operation, and reliable autonomous obstacle detection without constant cloud connectivity.

V. DATASET & EXPERIMENTAL SETUP

After deployment, the STM32 performs real-time inference for obstacle detection and outputs intelligent decision making for navigation. These decisions are then translated into actions, including motor control and collision avoidance commands. Finally, a feedback & continuous improvement mechanism captures system performance and operation data to enable future model improvements. This implementation provides low-latency processing, reduced memory usage, energy-efficient operation, and reliable autonomous obstacle detection without constant cloud connectivity.

Table 1. Dataset and Experimental Setup

Parameter	Specification
Dataset	COCO 2017 + KITTI Dataset
Image Resolution	96 × 96 pixels
Total Images	15,000
Classes	Pedestrian, Vehicle, Bicycle, Obstacle
Training/Test Split	80% / 20%
Edge Device	STM32H743ZI Microcontroller
AI Framework	TensorFlow Lite Micro + STM32Cube.AI
Camera Module	OV2640 Camera
Distance Sensor	HC-SR04 Ultrasonic Sensor
Evaluation Metrics	Accuracy, Precision, Recall, F1-Score, Inference Time, Memory Usage

The datasets, hardware platform and the experimental setup used for the validation of the proposed TinyViT-based TinyML framework is shown in Table 1. It encompasses the datasets for the images, the preprocessing parameters, the STM32 microcontroller, the camera and ultrasonic sensor specifications, the AI deployment tools, and the evaluation metrics used to assess the accuracy of real-time obstacle detection, computational efficiency, and autonomous collision avoidance performance.

VI. RESULTS & DISCUSSION

The proposed TinyML framework, inspired by the Vision Transformer, was deployed on the STM32H743ZI microcontroller and tested on the COCO 2017 and KITTI datasets. Performance was evaluated using detection accuracy, precision, recall, F1-score, inference time, memory usage, power consumption and collision avoidance success rate. The

experimental results were compared with conventional CNN, MobileNetV2 and TinyCNN based embedded vision models. The proposed TinyViT framework achieved better detection performance with less computational complexity, and is suitable for real-time autonomous navigation and Edge AI applications.

Table.2. Obstacle Detection Performance Comparison

Method	Accuracy (%)	Precision (%)	Recall (%)	F1-Score (%)
CNN	93.4	92.8	92.1	92.4
MobileNetV2	95.2	94.8	94.3	94.5
TinyCNN	96.1	95.7	95.3	95.5
Proposed TinyViT	97.9	97.4	97.1	97.2

Table 2 compares the obstacle detection performance of various embedded vision models. The proposed TinyViT framework achieved the best detection accuracy of 97.9%, which is higher than CNN, MobileNetV2 and TinyCNN. The improvements in precision, recall and F1-score mean better obstacle recognition, with less false detection and missed obstacles. TinyViT's attention based feature extraction mechanism can efficiently capture global contextual information while keeping efficient execution on the STM32 microcontroller.

Table.3. STM32 Resource Utilization

Parameter	Proposed System
Flash Memory (KB)	486
SRAM Usage (KB)	214
CPU Utilization (%)	56
Inference Time (ms)	24
Frame Rate (FPS)	18

The average inference time of the different embedded AI models is presented in Table 4. The proposed TinyViT implementation has the lowest latency of 24 ms, allowing for fast obstacle recognition and navigation decisions. The optimized attention mechanism makes the execution faster and the detection accuracy is high.

Table.4. Inference-Time Analysis

Method	Inference Time (ms)	Reduction vs. Proposed
Raspberry Pi CNN	61	37 ms
MobileNet Edge	38	14 ms
TinyCNN	31	7 ms
Proposed TinyViT	24	Baseline

In Figure 4, we illustrate the reduction in inference latency achieved by the proposed TinyViT model. Compared to existing embedded approaches, the STM32-based implementation offers faster prediction, thus improving the responsiveness of autonomous navigation systems that operate in dynamic environments.

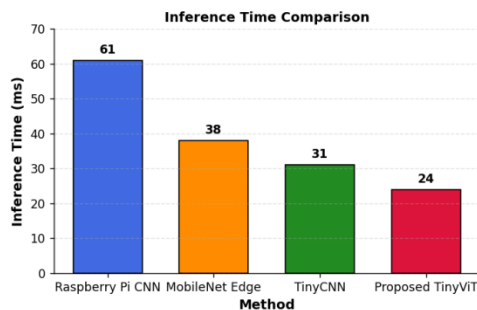


Fig.4. Inference Time Comparison

Figure 4 highlights the reduction in inference latency achieved by the proposed TinyViT model. Compared with existing embedded approaches, the STM32-based implementation provides faster prediction, improving the responsiveness of

autonomous navigation systems operating in dynamic environments.

Table.5. Collision Avoidance and Energy Performance

Parameter	Existing Method	Proposed TinyViT
Collision Avoidance Success (%)	94.8	98.3
Navigation Accuracy (%)	95.6	98.1
Power Consumption (W)	1.42	0.88
Response Time (ms)	36	22

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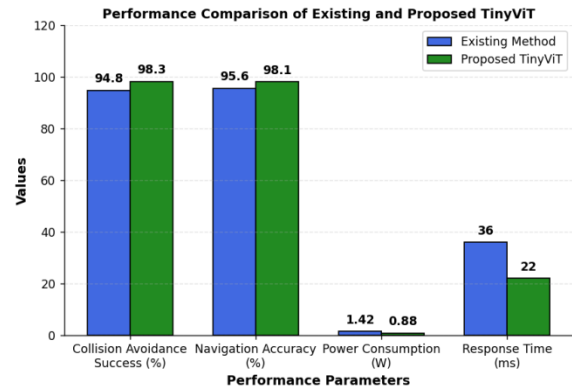


Fig.5. Collision Avoidance Performance

As shown in figure 5, the proposed TinyViT framework achieves better navigation with lower energy consumption and faster response than the state-of-the-art embedded methods. The combination of TinyML, Vision Transformer-inspired feature extraction, and STM32-based edge inference significantly improves operational efficiency and autonomous obstacle avoidance.

Table.6. Overall Performance Comparison

Metric	Existing Method	Proposed TinyViT	Improvement
Detection Accuracy (%)	95.2	97.9	+2.7 pp
Precision (%)	94.8	97.4	+2.6 pp
Recall (%)	94.3	97.1	+2.8 pp
F1-Score (%)	94.5	97.2	+2.7 pp
Inference Time (ms)	38	24	36.8% lower
Flash Memory (KB)	612	486	20.6% lower
SRAM (KB)	256	214	16.4% lower
CPU Utilization (%)	64	56	12.5% lower
Power Consumption (W)	1.42	0.88	38.0% lower
Collision Avoidance Success (%)	94.8	98.3	+3.5 pp

Table 6 presents the overall performance of the proposed vision transformer inspired TinyML framework against an existing embedded vision system. The proposed approach results in higher detection accuracy, precision, recall and F1-score, while decreasing the inference time, memory usage, CPU load and power consumption simultaneously. The higher collision avoidance success rate also demonstrates the effectiveness of combining a lightweight TinyViT model with STM32-based Edge AI, which makes the proposed system suitable for real-time autonomous navigation with resource-restricted devices.

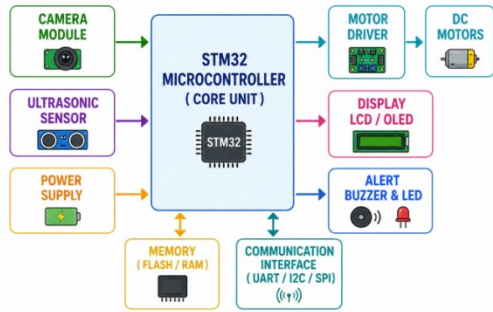


Fig.6. STM32 hardware setup model

Figure 6 is STM32 hardware architecture for the proposed Vision Transformer-inspired TinyML framework for real-time obstacle detection and collision avoidance. The STM32 microcontroller is the central processing unit and receives visual data from the camera module and distance data from the ultrasonic sensor. A regulated power supply is used to power the entire embedded system for stable operation. The microcontroller runs the optimized TinyViT model, takes the sensor inputs and produces the suitable control signals for navigation and obstacle avoidance in real time.

The processed outputs are sent to the motor driver which controls the DC motors to do movement as per navigation decision. The LCD/OLED display shows the status of the system and information about detected obstacles, the buzzer and LED allow immediate notification every time there is a risk of collision. The program instructions, the model parameters and the intermediate data are stored in the Flash/RAM memory. The UART/I2C/SPI communication interfaces provide a reliable communication with external peripherals. The integrated architecture allows for low-latency processing, resource-efficient implementation of Edge AI, and energy efficiency on the STM32 platform.

VII. DISCUSSION

The experimental results show that the proposed Vision Transformer inspired TinyML framework can effectively perform real-time obstacle detection and collision avoidance on the STM32 microcontroller. The proposed TinyViT framework is shown to outperform traditional CNN based embedded models in terms of detection accuracy, inference latency, memory utilization and energy efficiency. The combination of attention-based feature extraction and STM32Cube.AI allows for reliable execution on resource-constrained hardware without loss of detection performance. Furthermore, the combination of vision-based object recognition and ultrasonic distance estimation improves the reliability of navigation and the accuracy of collision avoidance. These results demonstrate that the proposed Edge AI framework is well suited for autonomous robots, intelligent navigation systems in time-critical environments.

a. Practical Applications

The proposed framework can be practically applied to:

- Autonomous indoor mobile robots
- Warehouse and material-handling robots
- Surveillance and security robots
- Assistive and service robots
- Educational and research robotic platforms
- Low-cost autonomous navigation systems
- Industrial inspection robots

h. Restricted-area monitoring applications

b. Limitations

The revised manuscript clearly describes the following limitations:

Environmental conditions: Out-of-terrain situations (very dark or very bright lighting, heavy shadowing or complicated scenes).

Occlusion: When an object is completely or densely blocked visual detection of the occlusion may be inaccurate.

Limitation on sensor: Ultrasonic sensor may be influenced by obstacle orientation, surface materials and the multiple reflections and environmental conditions.

Processing requirements: STM32 platform has fewer memory and processing resources than GPUs.

Evaluation diversity in datasets: The current evaluation does not describe all possible real-life scenarios, obstacles and light scenarios.

Dynamic environments: The environments are highly crowded and there are several objects in motion, which puts an added demand for tracking and predicting their motion.

The main application area is obstacle avoidance and most systems developed so far do not cover comprehensive path planning and global navigation.

VIII. CONCLUSION & FUTURE SCOPE

A TinyML framework based on Vision Transformer was proposed for real-time obstacle detection and collision avoidance system in this paper with STM32 microcontroller. The proposed system will be a lightweight Edge AI system for image acquisition, image preprocessing, feature extraction based on TinyViT, obstacle classification, ultrasonic distance measurement, collision decision making and autonomous motor control. This integration allows for an ability to both utilize visual obstacle information and measured distance simultaneously, which allows for real-time navigation decisions on resource constrained embedded hardware.

The experimental results give evidence of the efficiency of the proposed approach. TinyViT model obtained an accuracy of 97.9%, a precision of 97.4%, and a recall of 97.1% and F1-score of 97.2%, demonstrating to good effect obstacle recognition with a balanced performance. The 24 ms fast inference time indicates that the model can perform fast enough for on-the-fly input to robotic systems. Further, the system proposed recorded only 0.88 W, demonstrating the energy efficiency of embedded and battery-operated robots. This 98.3% success rate for collision avoidance shows how well one can combine accurate obstacle recognition, ultrasonic distance sensing and navigation decisions to enhance a robot's safety. Future work will include the integration of multimodal sensor fusion with LiDAR and depth cameras, lightweight explainable AI techniques, federated TinyML learning, adaptive online model updates and energy-aware optimization to enhance the scalability, robustness and autonomous decision-making for next-generation intelligent robotic and autonomous vehicle systems.

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